



**Politecnico
di Torino**

Politecnico di Torino

Nuclear fission plants

Assignment 2:

Thermal hydraulics in the hot subchannel of an AP1000 reactor

Master degree in Sustainable nuclear energy

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Chapter 1

Introduction and input data

The objective of this report is to perform a detailed thermal-hydraulic analysis of the hot subchannel in an AP1000 pressurized water reactor (PWR) under nominal operating conditions. The analysis focuses on the power profile and coolant thermodynamic evolution, the subcooled boiling and the radial thermal profiles, the critical heat flux and DNB analysis.

The calculation is carried out using the average values of heat flux and flow rate, assuming the hot channel closed and considering the hot-channel factors evaluated by nuclear calculations.

The AP1000 is a Generation III+ reactor designed by Westinghouse, characterized by the extensive use of passive safety systems and a simplified plant arrangement.

1.1 Geometric and Operating Parameters

The fundamental parameters for the subchannel modeling, including the geometric configuration of the 17×17 fuel assembly and the nominal thermal-hydraulic conditions, are summarized in Table 1.1.

From the input geometry of the square fuel lattice the quantities needed for the thermal-hydraulic subchannel analysis are derived as follows.

The inner cladding diameter follows directly from the cladding thickness:

$$D_{ci} = D_{co} - 2s \quad (1.1)$$

For a closed subchannel in a square array, the wetted perimeter is the rod circumference, while the coolant flow area is the unit-cell area minus the rod cross-section:

$$p_w = \pi D_{co} \quad A_c = w^2 - \frac{\pi}{4} D_{co}^2 \quad (1.2)$$

Table 1.1: Input parameters used in the thermal-hydraulic analysis [1], [2].

Symbol	Description	Value	Unit	SI value	SI unit
<i>Operating conditions</i>					
P_{nom}	Nominal thermal power	3400	MW	3.40×10^9	W
p_{sys}	System pressure	2250	psia	1.551×10^7	Pa
n_{rods}	Total number of fuel rods ^a	41 448	–	41 448	–
$\dot{m}_{\text{tot}}$	Total coolant mass flow rate	1.135×10^8	lbm/hr	1.430×10^4	kg/s
F_q	Heat flux hot-channel factor	2.6	–	2.6	–
f_{bp}	Bypass flow fraction	0.059	–	0.059	–
A_{flow}	Effective flow area	41.8	ft ²	3.883	m ²
T_{in}	Coolant inlet temperature	535	°F	279.4	°C
<i>Geometry (square 17×17 array)</i>					
D_{co}	Outer cladding diameter	0.374	in	9.500×10^{-3}	m
H_{active}	Active core height	168	in	4.267	m
w	Rod pitch	0.496	in	1.260×10^{-2}	m
s_{clad}	Cladding thickness	0.0225	in	5.715×10^{-4}	m
D_f	Fuel pellet diameter	0.3225	in	8.192×10^{-3}	m
R_{eq}	Equivalent core radius	119.7	in	3.040	m
<i>Auxiliary</i>					
T_{amb}	Ambient reference temperature	25	°C	25	°C
p_i	Internal fill-gas pressure	7	MPa	3×10^6	Pa
$\text{enrich}_{\text{avg}}$	Average fuel enrichment	3.38 %	$\frac{\text{kg}_{235-\text{U}}}{\text{kg}_{\text{U}_{\text{tot}}}}$	–	–

^a 157 fuel assemblies × 264 rods per assembly.

The equivalent diameter of the subchannel, used in the convective heat-transfer correlations, is then:

$$D_h = \frac{4 A_c}{p_w} \quad (1.3)$$

Finally, the fuel pellet cross-section and radius, used to convert the volumetric heat generation rate into linear and surface heat fluxes, are:

$$A_f = \frac{\pi}{4} D_f^2 \quad R_f = \frac{D_f}{2} \quad (1.4)$$

Chapter 2

Power profile and coolant thermodynamic evolution

This chapter describes the axial distribution of the fundamental parameters governing the thermal-hydraulic behavior of the hot subchannel. The analysis starts from the power distribution and proceeds to the evaluation of the coolant's thermodynamic state [3].

An axial discretization of the fuel element and subchannel is performed considering the active length (H) and placing the origin ($z = 0$) in the middle for simplicity.

2.1 Volumetric heat generation rate

At Beginning Of Life (BOL), the neutron flux distribution and therefore the power density distribution can be assumed to follow roughly a cosine behaviour axially and the Bessel function J_0 radially:

$$q_v(r, z) = q_{v,max} J_0\left(\frac{2.4048 r}{R_e}\right) \cos\left(\frac{\pi z}{H_e}\right) \quad (2.1)$$

Since the hot subchannel in the middle ($r = 0$) is considered, the equation becomes:

$$q_v(z) = q_{v,max} \cos\left(\frac{\pi z}{H_e}\right) \quad (2.2)$$

First, the average volumetric heat generation rate $\bar{q}_v$ is determined considering only a portion of the power generated inside the fuel, in this case 97.4%. Then the maximum volumetric heat generation is evaluated by multiplying this quantity by

the hot channel factor $F_q = 2.60$:

$$q_{v,max} = \bar{q}_v \cdot F_q \quad (2.3)$$

The extrapolated height H_e takes into account the transport mean free path of neutrons and the presence of a reflector:

$$H_e = H + 1.42 \lambda_{tr} + 2\delta \quad (2.4)$$

- Transport length: $\lambda_{tr} = \frac{1}{\Sigma_{tr}} = 0.29$ cm
- Reflector saving: $\delta = \frac{D_c}{D_R} L_R = 1.72$ cm
 - Diffusion coefficient in the core: $D_c = \frac{\lambda_{tr}}{3}$
 - Diffusion coefficient in the reflector: $D_R = 0.16$
 - Diffusion length in the reflector: $L_R = \sqrt{\frac{D_R}{\Sigma_a}} = 2.85$

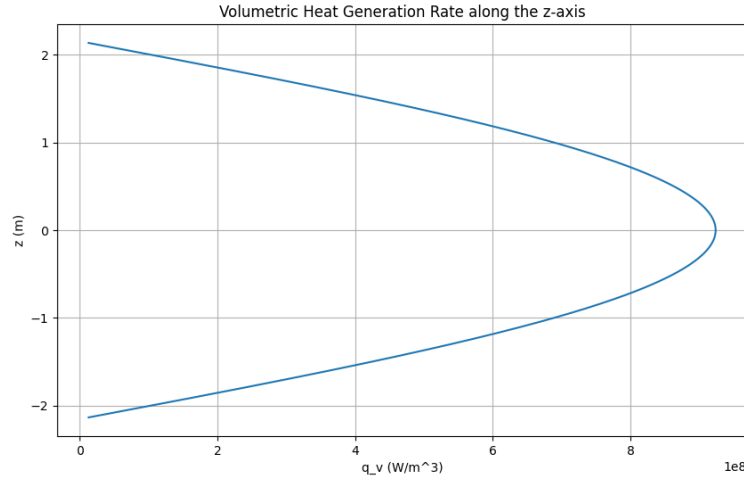


Figure 2.1: Volumetric Heat Generation Rate (q_v) along the axial coordinate z . The peak value $q_{v,max} \approx 9.2 \times 10^8$ W/m³ is reached at the core center ($z = 0$).

2.2 Average mass velocity in the channel

The total core mass flow rate $\dot{m}_{total}$ is corrected by a bypass factor to obtain the effective mass flow rate $\dot{m}_{eff}$. According to the design specifications:

$$\dot{m}_{eff} = \dot{m}_{total} \cdot (1 - f_{bp}) \quad (2.5)$$

where f_{bp} represents the fraction of the coolant that bypasses the active fuel region.

The average mass velocity G_{avg} is then calculated as the ratio between the effective mass flow rate and the total flow area of the core $A_{flow,tot}$:

$$G_{avg} = \frac{\dot{m}_{eff}}{A_{flow,tot}} \quad (2.6)$$

The calculated average mass velocity in the channel is $G_{avg} = 3465.36 \text{ kg}/(\text{m}^2 \text{ s})$.

2.3 Coolant specific enthalpy and temperature profiles

The coolant specific enthalpy profile is evaluated by integrating the axial power distribution. Following the energy balance for a subchannel with a chopped-cosine power distribution, the specific enthalpy $i(z)$ at a given axial position z is expressed as:

$$i(z) = i_{in} + \frac{A_f \cdot q_{v,max} \cdot H_e \cdot 1.0267}{\pi \cdot \dot{m}_{hc}} \left[\sin\left(\frac{\pi z}{H_e}\right) + \sin\left(\frac{\pi H_{active}}{2H_e}\right) \right] \quad (2.7)$$

where i_{in} is the coolant enthalpy at the inlet, A_f is the fuel cross-sectional area, and $\dot{m}_{hc}$ is the mass flow rate in the hot subchannel.

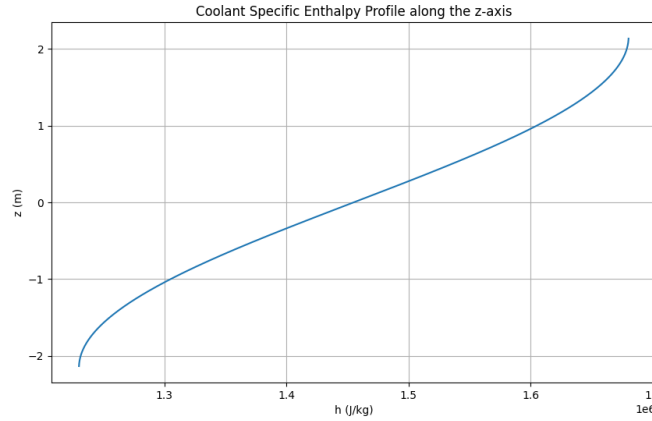


Figure 2.2: Coolant Specific Enthalpy Profile. The enthalpy rises from $i_{in} \approx 1.23 \times 10^6 \text{ J/kg}$ at the inlet to approximately $1.68 \times 10^6 \text{ J/kg}$ at the outlet.

The coolant temperature profile, $T_{cool}(z)$, is derived from the local specific enthalpy $i(z)$ and the constant system pressure p_{sys} . In this analysis, the temperature is calculated using the CoolProp library in Python.

A specific check is performed to account for the possibility of the coolant reaching saturation conditions within the hot subchannel. If the specific enthalpy is higher than the saturated liquid one, the temperature is limited to the saturation one.

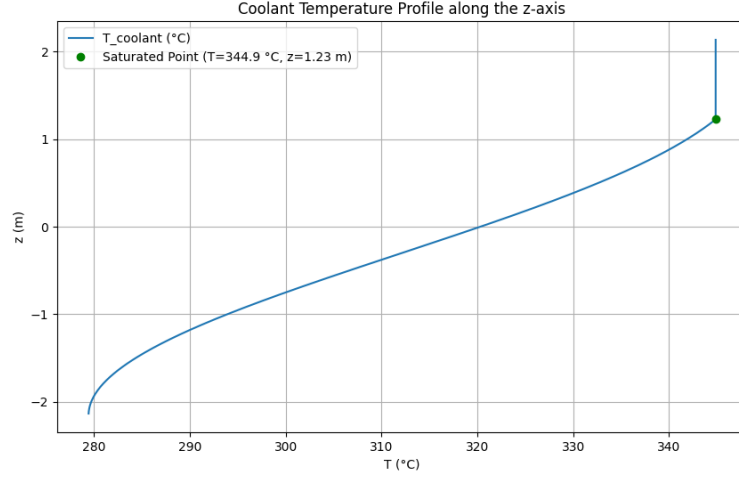


Figure 2.3: Coolant Temperature Profile along the z -axis.

As shown in Figure 2.3, the coolant enters at 279.4°C and reaches a saturation temperature of approximately 344.9°C near the outlet.

2.4 Equilibrium quality profile

To further characterize the thermodynamic state of the coolant along the subchannel, the equilibrium quality profile, $\chi_{eq}(z)$, is calculated. The equilibrium quality is a dimensionless parameter that relates the local specific enthalpy to the saturation properties at the system pressure:

$$\chi_{eq}(z) = \frac{i(z) - i_{l,sat}}{i_{v,sat} - i_{l,sat}} \quad (2.8)$$

where $i_{l,sat}$ and $i_{v,sat}$ are the specific enthalpies of saturated liquid and saturated vapor, respectively. In this analysis:

- A **negative value** ($\chi_{eq} < 0$) indicates that the coolant is a subcooled liquid. This is the condition observed in the lower and middle regions of the core.
- A **value of zero** ($\chi_{eq} = 0$) corresponds to the saturated liquid state, marking the point where the bulk coolant reaches T_{sat} .

- A **positive value** ($\chi_{eq} > 0$) would indicate a saturated mixture (saturated boiling).

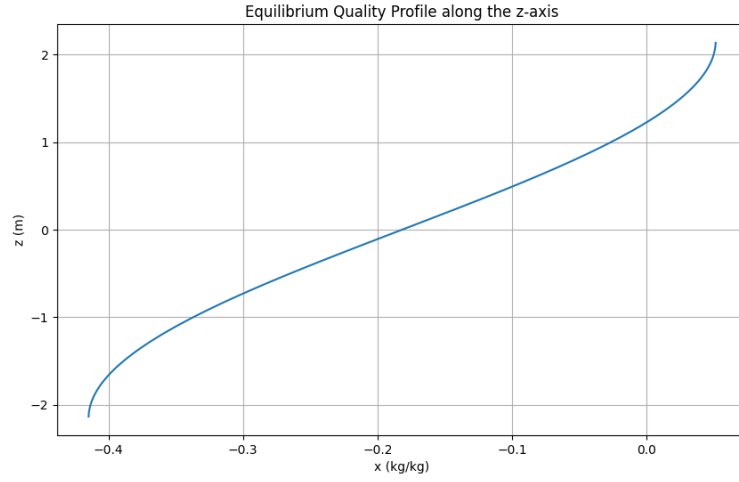


Figure 2.4: Equilibrium Quality Profile. The quality remains negative ($\chi_{eq} < 0$) for $z < 1.23$ m, indicating subcooled liquid, and reaches a maximum of ≈ 0.05 at the outlet.

The calculation is performed using the `PropsSI` function from the `CoolProp` library to ensure accurate evaluation of the saturation properties at 15.51 MPa (p_{sys}).

Chapter 3

Subcooled boiling and radial thermal profiles

This chapter is dedicated to the evaluation of the temperature profiles within the fuel element, starting from the previously calculated coolant temperature and proceeding inward to the centerline of the fuel pellet. The ultimate objective is to present the comprehensive thermal profiles, illustrating how temperatures vary along the radial direction [3].

3.1 Outer cladding wall temperature and subcooled boiling

In order to calculate the outer cladding temperature, a single-phase liquid coolant is assumed along the subchannel. This assumption allows the evaluation of the convective heat transfer coefficient by means of a corrected Dittus-Boelter correlation, which accounts for fully developed turbulent flow conditions. On the other hand, this hypothesis could underestimate the heat transfer coefficient h and overestimate the location of the boiling inception point.

$$T_{co} = T_{cool} + \frac{q_{flux}}{h_l} \quad (3.1)$$

$$Nu = C Re^{0.8} Pr^{0.4} = C \left(\frac{GD_h}{\mu} \right)^{0.8} \left(\frac{c_p \mu}{k} \right)^{0.4} \quad \text{and} \quad h_l = \frac{Nu k}{D_h} \quad (3.2)$$

The correction factor C strongly depends on the specific geometry, which in this study consists of a square array lattice, leading to:

$$C = 0.042 \frac{w}{D_{co}} - 0.024 \quad (3.3)$$

To account for subcooled nucleate boiling, the empirical Jens-Lottes correlation is applied:

$$T_{co,JL} = T_{sat} + 25 q^{0.25} e^{-\frac{p}{62}} \quad \text{with } [p] = \text{bar}, [T] = ^\circ \text{C}, [q] = \frac{MW}{m^2} \quad (3.4)$$

Finally, the actual outer cladding temperature profile is obtained by combining the single-phase and the boiling curves as illustrated in Figure 3.1. The selection of the minimum value between the single-phase liquid temperature ($T_{co,SP}$) and the subcooled nucleate boiling curve ($T_{co,JL}$) mathematically models the physical transition between different heat transfer regimes. The system naturally operates along the path of minimum thermal resistance. Consequently, single-phase convection dominates in the lower region of the subchannel, as the coolant moves upward and the heat flux rises, the cladding surface temperature triggers the Onset of Nucleate Boiling (ONB). This phase change mechanism provides an exceptionally high heat transfer coefficient that effectively clamps the cladding wall temperature slightly above the saturation threshold (T_{sat}), leading to the final profile.

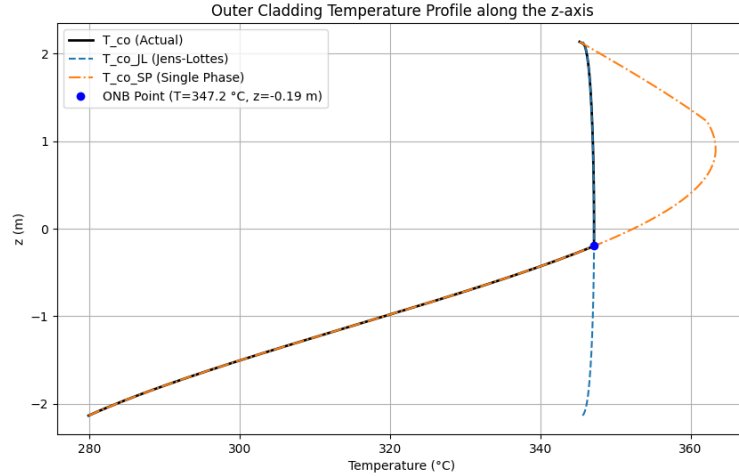


Figure 3.1: Axial temperature profile of the outer cladding surface compared with single-phase and Jens-Lottes predictions.

3.1.1 Bubble detachment (Griffith)

Following the ONB, bubbles begin to nucleate at the wall, increasing the void fraction (initially zero). As they grow larger, detachment occurs and the bubbles enter the flow stream region.

The Griffith model can be adopted to calculate the liquid temperature at detachment:

$$T_{det} = T_{sat} - \frac{q_{flux}}{5h_l} \quad (3.5)$$

Table 3.1: Axial location and coolant temperature of the boiling milestones along the hot subchannel.

Point	Axial position z (m)	Coolant temperature (°C)
ONB	−0.19	315.2
Bubble detachment	+0.90	340.4
Saturation	+1.23	344.9

At ONB the outer cladding wall temperature is $T_{co} = 347.2$ °C.

3.1.2 Flow quality profile (Bowring–Rouhani model)

Due to the definition itself of flow quality, it is null before detachment:

$$\chi_f = \frac{\dot{m}_v}{\dot{m}_v + \dot{m}_l} \quad (3.6)$$

where $\dot{m}_v$ is the vapor mass flow rate and $\dot{m}_l$ is the liquid mass flow rate. After detachment, the flow quality can be evaluated with the Bowring-Rouhani model.

Heat is exchanged at the heated surface by several mechanisms:

$$q = q_e + q_a + q_{sp} + q_{cond} \quad (3.7)$$

- q_e : Latent heat of evaporation;
- q_a : Convection caused by bubble agitation;
- q_{sp} : Single-phase heat transfer between patches of bubbles, equal to zero only considering fully developed boiling at the bubble detachment, otherwise if partial boiling is present:

$$q_{sp} = h_l(T_{sat} - T_{cool}) \quad (3.8)$$

- q_{cond} : Recondensation of the bubbles at the wall, negligible with respect to the others under the Bowring model (demonstrated experimentally).

The ratio between latent and convective heat is defined as ε . It can be evaluated as:

1. Rouhani:

$$\varepsilon = \frac{\rho_l}{\rho_{g,sat} H_{fg}} (i_{l,sat} - i_l) \quad (3.9)$$

2. Bowring:

$$\varepsilon = 1.3 \quad \text{for } 10 < p < 50 \left[\frac{\text{kg}_f}{\text{cm}^2} \right] \quad \text{or} \quad \varepsilon = 1.6 \quad \text{for } p > 50 \left[\frac{\text{kg}_f}{\text{cm}^2} \right] \quad (3.10)$$

The actual quality in the subcooled region is:

$$\chi = \int_{z_d}^z \frac{q_e p_{wet}}{H_{fg} W} dz = \int_{z_d}^z \frac{p_{wet} (q - q_{sp})}{H_{fg} W (1 + \varepsilon)} dz \quad (3.11)$$

where H_{fg} is the latent heat of vaporization, and $W = G_{avg} A_{cross}$.

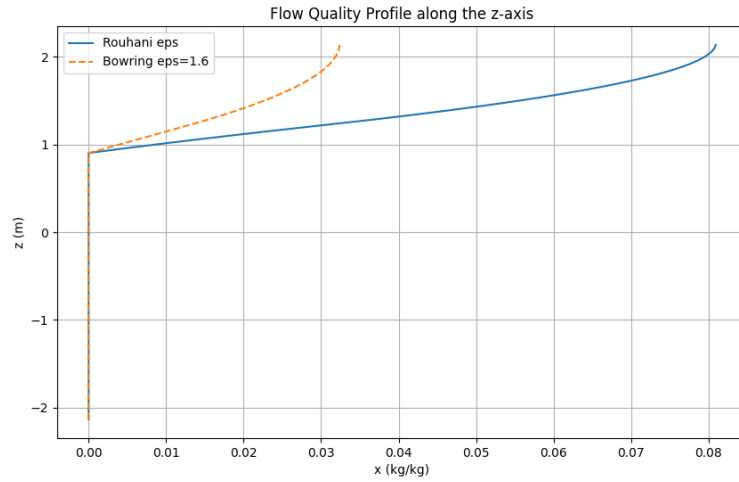


Figure 3.2: Flow quality with Rouhani and Bowring methods.

3.1.3 Void fraction profile

In a similar manner to the flow quality, the void fraction remains zero before the ONB. Beyond this point, two distinct regions can be identified:

1. **ONB-Detachment:** the void fraction exhibits a linear behavior, and its value at the detachment point can be estimated using the Maurer correlation:

$$\alpha = \frac{4\delta}{D_h} \quad \text{with} \quad \delta = 0.0666 R_d \quad (3.12)$$

The bubble layer thickness, δ , is evaluated via the Bowring method, which assumes hemispherical bubbles of radius R_d covering 50% of the wall surface. The bubble radius at detachment follows the Rouhani correlation:

$$R_d = \frac{2.37 \cdot 10^{-3}}{p^{0.237}} \quad \text{with} \quad [p] = \text{bar} \quad (3.13)$$

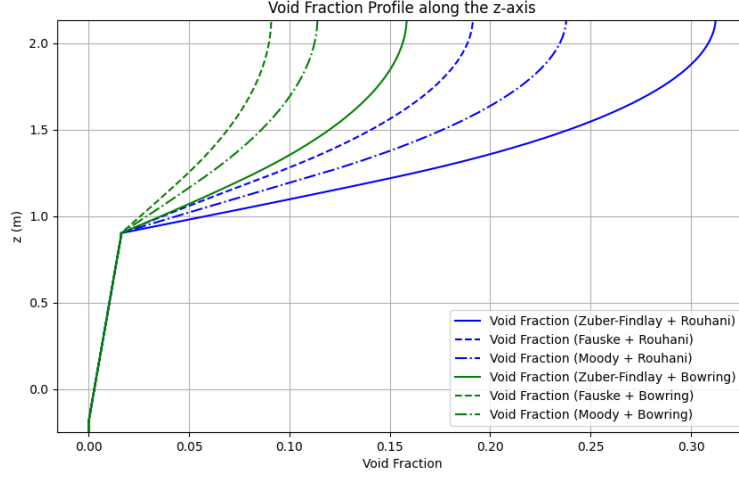


Figure 3.3: Axial evolution of the void fraction from the ONB point to the channel exit, comparing different slip-ratio and drift-flux correlation models.

2. **Detachment to subchannel outlet:** the total void fraction is obtained by adding the contribution of the chosen correlation to the value previously calculated at the detachment point. Specifically, α can be evaluated using different models:

- Zuber-Findlay:

$$\alpha = \frac{\chi / \rho_g}{C_0 \left(\frac{\chi}{\rho_g} + \frac{1-\chi}{\rho_l} \right) + C_1 \frac{1}{G} \left[\sigma g \frac{(\rho_l - \rho_g)}{\rho_l^2} \right]^{0.25}} \quad (3.14)$$

with constants $C_0 = 1.13$ and $C_1 = \frac{1.18+1.4}{2}$, selected from the Collier-Thome theory for bubbly flows, and σ is the surface tension.

- Slip-ratio model:

$$\alpha = \frac{\chi}{\chi + (1 - \chi) \frac{\rho_g}{\rho_l} S} \quad (3.15)$$

where the slip ratio S can be evaluated as:

$$\text{Fauske: } \left(\frac{\rho_l}{\rho_g} \right)^{1/2} \quad \text{Moody: } \left(\frac{\rho_l}{\rho_g} \right)^{1/3}$$

Figure 3.3 illustrates the axial evolution of the void fraction, showing the transition from the linear trend in the subcooled region to the fully developed boiling zone. The comparison highlights the differences between the drift-flux and slip-ratio models in predicting the vapor distribution along the channel.

3.2 Inner cladding wall temperature with Zircaloy thermal conductivity

The inner cladding temperature is computed by considering heat conduction through a hollow cylinder, accounting for the variation of the cladding thermal conductivity with temperature:

$$k_{cl} = 11.45 + 1.425 \cdot 10^{-2} T \quad [k_{cl}] = \frac{\text{W}}{\text{m} \cdot ^\circ \text{C}} \quad (3.16)$$

To obtain the temperature profile, the following integral equation is solved analytically:

$$\int_{T_{co}}^{T_{ci}} k_{cl} dT = \frac{q_v A_f}{2\pi} \ln \left(\frac{D_{co}}{D_{ci}} \right) \quad (3.17)$$

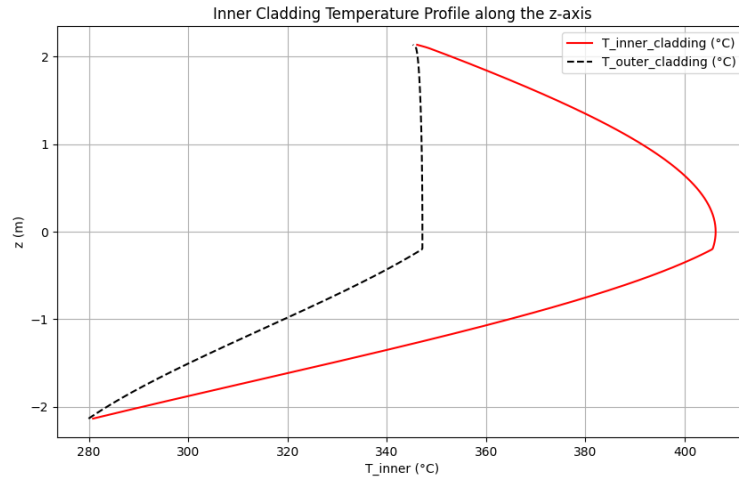


Figure 3.4: Axial distribution of the inner cladding temperature (T_{ci}) compared to the outer cladding temperature (T_{co}).

3.3 Fuel pellet surface temperature

To compute the fuel pellet surface temperature, the presence of the gap between the pellet and the cladding, which is initially filled with Helium, must be considered. The pellet surface temperature is given by:

$$T_{f,s}(z) = T_{ci}(z) + \frac{q_v(z)A_f}{\pi D_f h_g^T} \quad (3.18)$$

where h_g^T is the total gap conductance, which is composed of three parallel heat transfer mechanisms:

$$h_g^T = h_{rad} + h_{gap} + h_c \quad (3.19)$$

- **Radiative term** (h_{rad}): it is evaluated by assuming the emissivity of both the fuel and cladding to be equal to unity, thereby obtaining a conservative maximum value.

$$h_{rad} = \sigma \frac{T_{f,s}^4 - T_{ci}^4}{T_{f,s} - T_{ci}} \quad \text{with} \quad \sigma = 5.67 \cdot 10^{-8} \frac{\text{W}}{\text{m}^2 \cdot \text{K}^4} \quad (3.20)$$

- **Gap component** (h_{gap}): it represents a series of thermal resistances, specifically the resistance due to the presence of the gas fill and the heat exchange resistances at the solid surfaces. Since the former is highly dominant, the surface jump terms can be neglected. Consequently, it can be computed via the Ross and Stoute correlation:

$$h_{gap} = \frac{k_{gas}}{\beta_2(R_f + R_{cl}) + l_f + l_{cl} + \delta} \quad \text{with} \quad \beta_2(R_f + R_{cl}) + l_f + l_{cl} = 2.54 \cdot 10^{-3} \text{ cm} \quad (3.21)$$

where δ is the effective gap thickness. Since the fill gas inside the cladding is Helium, its thermal conductivity is evaluated as:

$$k_{He} = A \cdot T^N \quad \text{with} \quad A = 0.1763 \cdot 10^{-2} \quad \text{and} \quad N = 0.77163 \quad (3.22)$$

- **Contact term** (h_c): it must be evaluated only in the event of physical contact between the fuel pellet and the cladding caused by differential thermal expansion.

The gap thickness is not constant over the active height, as it depends on two main phenomena:

1. **Thermal expansion**: due to the temperature increase, both the cladding and the fuel pellet expand. The variation in diameter can be evaluated as:

$$\Delta D = D^{T_a} \alpha (\bar{T} - T_a) \quad (3.23)$$

where T_a is the ambient temperature set at 25 °C, $\bar{T}$ is the average temperature of the component under analysis, and α is the linear thermal expansion coefficient, which depends on the material:

$$\begin{cases} \text{Fuel (UO}_2\text{):} & \alpha_f = 7.87 \cdot 10^{-6} + 3.9 \cdot 10^{-9} \bar{T}_f \\ \text{Cladding (Zircaloy):} & \alpha_{cl} = 5.62 \cdot 10^{-6} + 3.162 \cdot 10^{-9} \bar{T}_{cl} \end{cases} \quad (3.24)$$

Cladding thermal expansion represents a positive contribution to the final gap thickness, whereas fuel pellet expansion provides a negative contribution.

2. **Cladding elastic deformation:** the radial deformation due to pressure is evaluated by modeling the cladding as an infinite thick-walled cylinder:

$$\frac{\Delta r_{ci}}{r_{ci}} = \frac{1}{E} \frac{1}{\gamma^2 - 1} \left\{ p_i \left[(1 - \nu) + (1 + \nu)\gamma^2 \right] - 2\gamma^2 p_{sys} \right\} \quad (3.25)$$

where $\gamma = \frac{r_{co}}{r_{ci}}$, p_i is the internal fill-gas pressure, $\nu = 0.43$, and E is the Young's modulus of the Zircaloy cladding. The Young's modulus depends linearly on the cladding temperature:

$$E = 1.148 \cdot 10^{11} - 5.99 \cdot 10^7 T \quad \text{with} \quad [T] = \text{K} \quad (3.26)$$

Since the system pressure (p_{sys}) is higher than the internal pressure (p_i), this elastic deformation provides a negative contribution to the gap thickness.

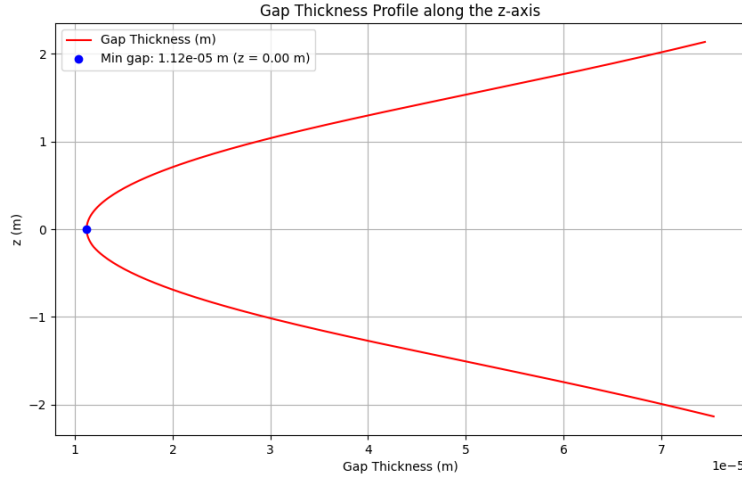


Figure 3.5: Axial profile of the effective gap thickness along the active core height.

Since the fuel properties depend heavily on temperature, an iterative procedure must be established. An initial guess for the pellet surface temperature is chosen

based on the cladding inner temperature, namely $T_{f,S}^{(0)} = T_{ci} + 100$ °C. To evaluate the average fuel temperature $\bar{T}_f$, a parabolic profile is assumed inside the pellet (modeled as a solid cylinder with uniform volumetric heat generation), so that the centerline and the mean temperatures can be expressed as:

$$T_{f,C} = T_{f,S} + \frac{q_v A_f}{4\pi k_{\text{UO}_2}}, \quad \bar{T}_f = T_{f,S} + \frac{q_v A_f}{8\pi k_{\text{UO}_2}} \quad (3.27)$$

The thermal conductivity of the UO_2 pellet depends on temperature and is evaluated through the Westinghouse correlation as a function of $\bar{T}_f$:

$$k_{\text{UO}_2}(\bar{T}_f) = \frac{1}{11.8 + 0.0238 \bar{T}_f} + 8.775 \cdot 10^{-13} \bar{T}_f^3 \quad [\text{W}/(\text{cm} \cdot \text{K})] \quad (3.28)$$

At each iteration k , the updated fuel temperatures are used to recompute k_{UO_2} , and, together with the corresponding gap conductance h_g^T , the pellet surface temperature is updated as:

$$T_{f,S}^{(k+1)} = T_{ci} + \frac{q_v A_f}{\pi D_f h_g^{T,(k)}} \quad (3.29)$$

The loop is repeated until the relative variation of the pellet surface temperature falls below a prescribed convergence tolerance. At convergence, it is also verified that the effective gap thickness remains positive along the entire active length, thereby confirming that no pellet-cladding contact occurs.

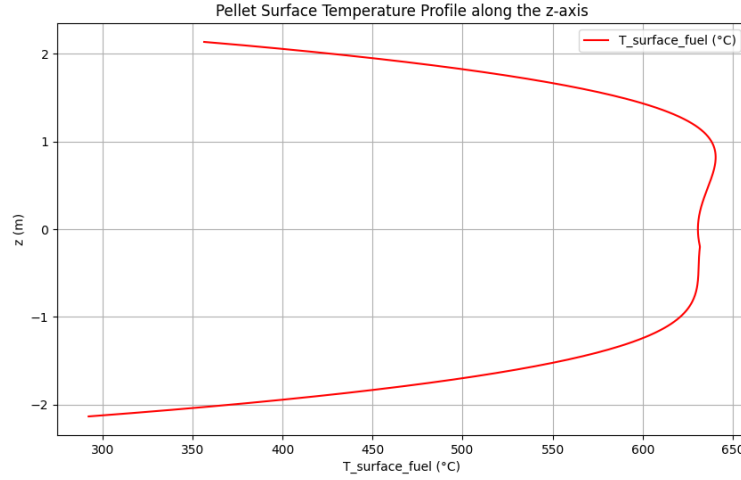


Figure 3.6: Converged axial temperature profiles for fuel pellet surface ($T_{f,S}$).

3.4 Fuel pellet centerline temperature

The fuel centerline temperature is computed by considering heat conduction within a solid cylinder, neglecting axial conduction. The governing differential equation is given by:

$$\frac{d^2T}{dr^2} + \frac{1}{r} \frac{dT}{dr} + \frac{q_v f_{rob}}{k_{\text{UO}_2}} = 0 \quad (3.30)$$

subject to the boundary conditions $\left. \frac{dT}{dr} \right|_{r=0} = 0$ (symmetry) and $T(R_{\text{pellet}}) = T_{f,S}(z)$.

The volumetric heat source is corrected by the Robertson factor $f_{rob} \approx 0.965$ to account for the centerline flux depression. This factor depends on the pellet diameter $D_f = 0.3225$ inches and the average fuel enrichment $e_{\text{mean}} = 3.38\%$ (evaluated as the mean value across the three enrichment regions).

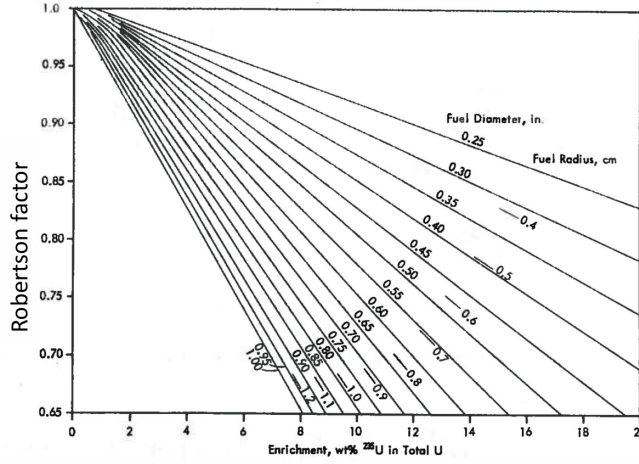


Figure 3.7: The Robertson factor chart used to determine the centerline flux depression correction.

The pellet radius is discretized using N_r equally spaced nodes. A conservative finite-difference scheme is adopted to numerically solve the governing equation:

$$\frac{r_{i+1/2} k_{i+1/2} (T_{i+1} - T_i) - r_{i-1/2} k_{i-1/2} (T_i - T_{i-1})}{\Delta r^2} = -q_v(z) r_i \quad (3.31)$$

where the face-centered radii $r_{i\pm 1/2} = r_i \pm \Delta r/2$ and the face conductivities $k_{i\pm 1/2} = \frac{1}{2}(k_i + k_{i\pm 1})$ are implemented to preserve flux continuity across adjacent cells. The two boundary conditions are imposed algebraically as $T_0 = T_1$ and $T_{N_r-1} = T_{f,S}(z)$.

At each axial section, the resulting tridiagonal linear system $\mathbf{A} \mathbf{T} = \mathbf{b}$ is assembled and solved directly. Since k_{UO_2} depends heavily on the unknown temperature

field, the system is non-linear and must be treated iteratively. At iteration k , the matrix is constructed using $k_{\text{UO}_2}(T^{(k)})$, and the updated temperature field $T^{(k+1)}$ is computed via an under-relaxation scheme:

$$T^{(k+1)} = \alpha T_{\text{calc}}^{(k+1)} + (1 - \alpha) T^{(k)}, \quad \alpha = 0.5 \quad (3.32)$$

The convergence loop terminates when the condition $\max_i |T_i^{(k+1)} - T_i^{(k)}| < 10^{-6}$ is satisfied. The pellet centerline temperature at a given axial position is finally recovered from the innermost node, $T_{f,C}(z) = T_0(z)$.

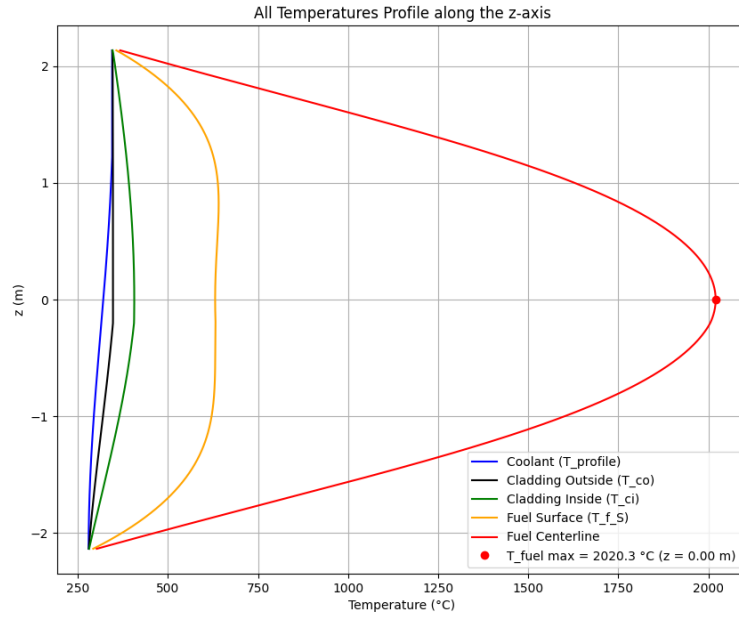


Figure 3.8: Axial temperature profiles computed across the hotspot subchannel, mapping the thermal gradient from the coolant bulk up to the fuel pellet centerline.

Figure 3.8 presents the comprehensive axial temperature distributions from the coolant to the fuel centerline. The results clearly illustrate how the maximum temperature is achieved at the core midplane due to the chopped-cosine power shape, always remaining under the melting temperature of the fuel $T_{\text{melt},\text{UO}_2} \approx 2800^\circ\text{C}$.

Chapter 4

Critical heat flux and DNB analysis

This chapter focuses on the analysis of the Critical Heat Flux (CHF) and Departure from Nucleate Boiling (DNB), which represent some of the most important thermo-hydraulic safety aspects in nuclear reactors. Particular attention is devoted to the prediction of the critical heat flux along the fuel channel and to the evaluation of the minimum DNBR location, since these quantities are directly related to the thermal safety margin of the reactor core [4].

4.1 Critical heat flux: W-3 correlation, grid spacer factor and Tong correction

The Critical Heat Flux (CHF) is the maximum heat flux that can be removed from a heated surface while maintaining effective cooling through nucleate boiling. When this limit is exceeded, a “boiling crisis” occurs, namely a sudden deterioration of heat transfer that causes a rapid increase in the wall temperature, which may lead to overheating and damage of the fuel cladding. For this reason, in nuclear reactors the CHF defines the thermo-hydraulic safety limit of the fuel.

The uniform heat flux is first evaluated using the W-3 correlation by Tong. Since this correlation is used for 15×15 fuel assemblies, it is adapted to the present 17×17 configuration considering a correction factor of 0.88.

The W-3 correlation is:

$$\begin{aligned}
 q_c = & \{ (2.022 - 0.06238 p) + (0.1722 - 0.01427 p) \exp [(18.177 - 0.5987 p) \chi_c] \} \\
 & \cdot [(0.1484 - 1.596 \chi_c + 0.1729 \chi_c |\chi_c|) 2.326 G + 3271] \\
 & \cdot (1.157 - 0.869 \chi_c) \\
 & \cdot [0.2664 + 0.8357 \exp(-124.1 D_h)] \\
 & \cdot [0.8285 + 0.0003413 (i_{l,sat} - i_{in})] \\
 [q_c] = & \frac{\text{kW}}{\text{m}^2} \quad [p] = \text{MPa}
 \end{aligned}$$

where χ_c , G , D_h , $i_{l,sat}$ and i_{in} are the quantities already introduced in the previous chapters.

This correlation does not take into account the presence of spacer grids that promote turbulence, improving the heat exchange (higher CHF). So W-3 is adapted by multiplying it by the grid spacer factor F_s :

$$F_s = \left(\frac{p}{225.896} \right)^{0.5} (1.445 - 0.0371 L) \exp \left(\frac{\chi + 0.2}{2} \right) \left[-0.73 + K_s \left(\frac{G}{10^6} \right)^{\alpha/0.019} \right]^{0.35}$$

$$[p] = \text{psi} \quad [L] = \text{ft} \quad [G] = \frac{\text{lb}}{\text{ft}^2 \text{hr}}$$

In this expression the quantities are no longer in SI units. Here L is the heated length (active height), $\alpha = 0.038$, and K_s is the axial grid spacing coefficient, which depends on the grid spacing $L_s = \frac{L}{N_{grid}}$ (in inches, with $N_{grid} = 10$) and is extrapolated from the table below.

L_s (in)	32	26	20
K_s	0.027	0.046	0.066

Table 4.1: Axial grid spacing coefficient values.

So the final equation of the uniform heat flux is:

$$q_{c,EU} = 0.88 q_c F_s$$

In real conditions the critical heat flux is slightly overpredicted with the uniform heat flux. For this reason Tong developed a correction factor F that provides the non-uniform heat flux:

$$q_{c,NU} = \frac{q_{c,EU}}{F}$$

The correction factor F is evaluated in this way:

$$F = \frac{C \int_{L_{ONB}}^{L_{c,NU}} q(z') \exp[-C(L_{c,NU} - z')] dz'}{q_{local} [1 - \exp(-C \cdot L_{c,NU})]}$$

$$C = 0.15 \frac{[1 - \chi(L_{c,NU})]^{4.31}}{G^{0.478}} \quad [z] = \text{in} \quad [G] = \frac{\text{Mlb}}{\text{ft}^2 \text{hr}}$$

This Tong correction is used starting from the ONB point and then, for each axial position z_i , it evaluates how the previous heat flux distribution affects the local CHF.

4.2 DNBR, minimum DNBR and its axial location

The DNBR (Departure from Nucleate Boiling Ratio) is the ratio between the available critical heat flux and the actual operating heat flux q .

$$DNBR = \frac{q_{c,NU}}{q}$$

It is the main thermo-hydraulic safety parameter in PWRs, since it measures the safety margin with respect to the onset of Departure from Nucleate Boiling (DNB).

The most important parameter is the MDNBR, namely the minimum value of DNBR along the channel, which identifies the most thermally critical location within the reactor core. In this case the minimum DNBR is 1.86 at the position $z = 0.91$ m, so this value satisfies the imposed limit (MDNBR = 1.85).

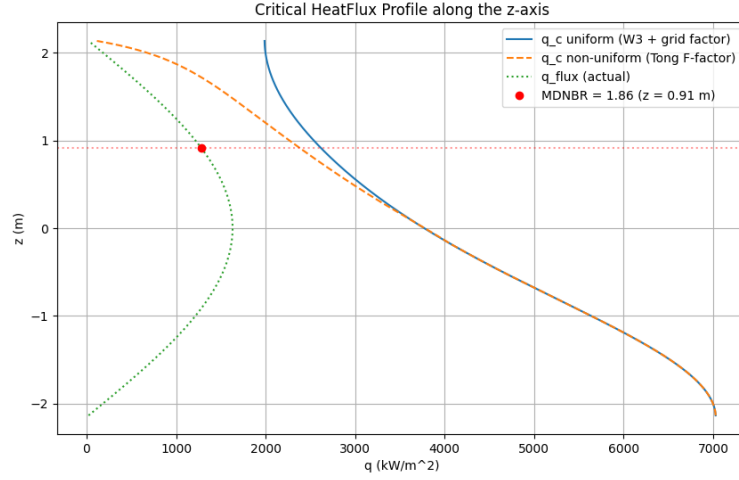


Figure 4.1: Axial evaluation of the critical heat flux and location of the MDNBR.

Finally, the thermal analysis confirms that the design thermal limits are respected. These are summarized in the table below.

Parameter	Computed Value	Design Limit	Status
MDNBR	1.86	≥ 1.85	OK
Axial position of MDNBR	+0.91 m	—	—
Maximum fuel centerline temperature	2020.3 °C	≤ 2800 °C	OK
Maximum cladding outer temperature	347.2 °C	≤ 350 °C	OK

Table 4.2: Verification of thermal design limits.

The design limits are taken from the reactor manual, while the cladding temperature limit was selected according to typical PWR thermo-hydraulic design criteria.

Chapter 5

Conclusions

This report carried out a detailed thermal-hydraulic analysis of the hot subchannel of an AP1000 reactor under nominal conditions, following the heat path from the chopped-cosine power profile up to the fuel pellet centerline and assessing the resulting safety margins.

The coolant enters at 279.4°C and reaches saturation at approximately 344.9°C in the upper region, with the onset of subcooled nucleate boiling predicted at $z = -0.19$ m. The radial reconstruction through cladding, gap and pellet yields a maximum fuel centerline temperature of 2020.3°C at the core midplane, consistently with the peak of the power shape. The CHF analysis, based on the W-3 correlation corrected for the 17×17 geometry with the grid spacer factor and the Tong correction, provides the most stringent indicator: the minimum DNBR is 1.86 at $z = 0.91$ m, located in the upper portion of the channel rather than at the midplane. All the verified parameters, summarized in Table 4.2, satisfy the imposed design limits.

These results were obtained under conservative assumptions: the single-phase hypothesis for the outer cladding overestimates the boiling inception point, while the unit emissivity in the radiative gap term provides a conservative upper bound on temperature. The fact that the design limits, and in particular the minimum DNBR, are met despite these choices reinforces the robustness of the predicted safety margins. The analysis returns a physically coherent picture, in which the power shape drives the peak temperatures toward the midplane while the most critical condition for the boiling crisis is shifted downstream.

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